Cameroon Electronic Health Record (EHR) Program – Planning Phase Blueprint

Prepared for: Founding Team and Stakeholders

Executive Summary

This document defines the planning phase for a national-scale Electronic Health Record (EHR) initiative in Cameroon. The primary objective is to understand healthcare workflows, regulatory requirements, interoperability needs, public health reporting requirements, and organizational readiness before software development begins. The planning phase emphasizes workflow discovery, stakeholder engagement, compliance, governance, interoperability through FHIR, and creation of a future de-identified public health data lake.

Vision

Build a modern, FHIR-first, clinician-centered EHR platform that supports hospitals, clinics, laboratories, imaging departments, pharmacies, administrators, and public health agencies. The platform should improve patient care, support operational efficiency, strengthen reporting, and enable evidence-based public health decision making.

Guiding Principles

1. Workflow before software. 2. Clinical safety first. 3. Privacy and security by design. 4. FHIR-first interoperability. 5. Mobile-first accessibility. 6. Low-bandwidth resilience. 7. Scalability across Cameroon. 8. Data-driven public health. 9. Standards-based architecture. 10. Sustainable local ownership.

Planning Phase Objectives

Map healthcare workflows; document forms and data elements; understand ministry requirements; identify compliance obligations; establish governance; define master data; identify interoperability needs; design public health reporting strategy; develop product roadmap; create implementation strategy.

Stakeholder Engagement

Engage hospital executives, nurses, physicians, pharmacists, laboratory personnel, imaging staff, medical records staff, finance teams, IT personnel, Ministry of Health representatives, public health agencies, regional health offices, private hospitals, teaching institutions, and patient representatives.

Hospital Workflow Assessment

Select a pilot hospital. Conduct observations in registration, emergency, outpatient services, inpatient units, nursing operations, physician workflows, pharmacy, laboratory, imaging, operating theaters, maternity, billing, inventory, and discharge processes. Follow real patient journeys and document every decision point, handoff, delay, form, and communication method.

Nursing Workflow Analysis

Admission assessment, triage, vital signs collection, medication administration, wound care, care plans, shift handoff, patient education, discharge planning, documentation burden, escalation workflows, and reporting requirements.

Physician Workflow Analysis

Patient assessment, order entry, consult requests, progress notes, procedure documentation, discharge summaries, referrals, diagnosis coding, and follow-up management.

Departmental Analysis

Document workflows for laboratory, radiology, pharmacy, surgery, emergency, maternity, intensive care, outpatient clinics, rehabilitation, billing, supply chain, and medical records.

Regulatory and Compliance Assessment

Meet with Ministry of Health officials to gather policies, regulations, record retention requirements, reporting mandates, privacy obligations, licensing considerations, national program requirements, health information exchange objectives, and digital health priorities.

Public Health Integration Strategy

Meet public health authorities to understand disease surveillance, outbreak management, vaccination reporting, maternal health reporting, tuberculosis programs, HIV programs, malaria monitoring, mortality reporting, and regional analytics requirements.

De-Identified National Data Lake Concept

Create a secondary analytics environment that excludes patient identifiers while retaining meaningful healthcare indicators. Potential analytics domains include disease prevalence, utilization trends, mortality analysis, vaccination coverage, maternal health outcomes, chronic disease surveillance, geographic distribution, and resource utilization.

Data Standards and Interoperability

Define a national healthcare data dictionary. Map future system entities to FHIR resources such as Patient, Encounter, Practitioner, Observation, MedicationRequest, DiagnosticReport, AllergyIntolerance, Procedure, Condition, CarePlan, and Immunization.

Data Governance

Define ownership, stewardship, retention schedules, audit requirements, access controls, consent policies, data quality processes, and escalation procedures.

Security and Privacy Planning

Perform threat modeling; define identity management requirements; establish audit logging principles; define encryption standards; establish backup and disaster recovery requirements; determine hosting constraints and operational responsibilities.

Infrastructure Assessment

Evaluate internet connectivity, electrical reliability, device availability, local IT support capability, server requirements, cloud readiness, cybersecurity maturity, and networking capacity across facilities.

Requirements Collection Deliverables

Process maps, workflow diagrams, requirements catalog, forms inventory, data dictionary, reporting inventory, interoperability inventory, public health reporting framework, compliance documentation, and executive recommendations.

Product Roadmap

Phase 1: Registration, scheduling, clinical documentation, pharmacy, laboratory. Phase 2: Inpatient workflows, medication administration, billing, inventory. Phase 3: Interoperability, patient portal, mobile applications. Phase 4: AI-assisted workflows, clinical decision support, analytics, predictive models.

Project Governance

Establish steering committee, clinical advisory council, technical architecture board, security committee, and implementation working groups.

Success Metrics

Workflow documentation completeness, stakeholder participation, regulatory alignment, data model readiness, interoperability readiness, public health integration readiness, executive approval, and pilot implementation readiness.

Planning Phase Timeline

Months 1-2: Discovery and stakeholder engagement. Months 3-4: Workflow mapping and requirements gathering. Month 5: Regulatory and public health analysis. Month 6: Data architecture, governance, roadmap, and final recommendations.

Final Recommendation

No software development should begin until workflow mapping, regulatory assessments, stakeholder interviews, governance planning, interoperability strategy, and public health integration planning are completed and approved. The outcome of this phase should be a validated blueprint that guides all future design and implementation decisions.